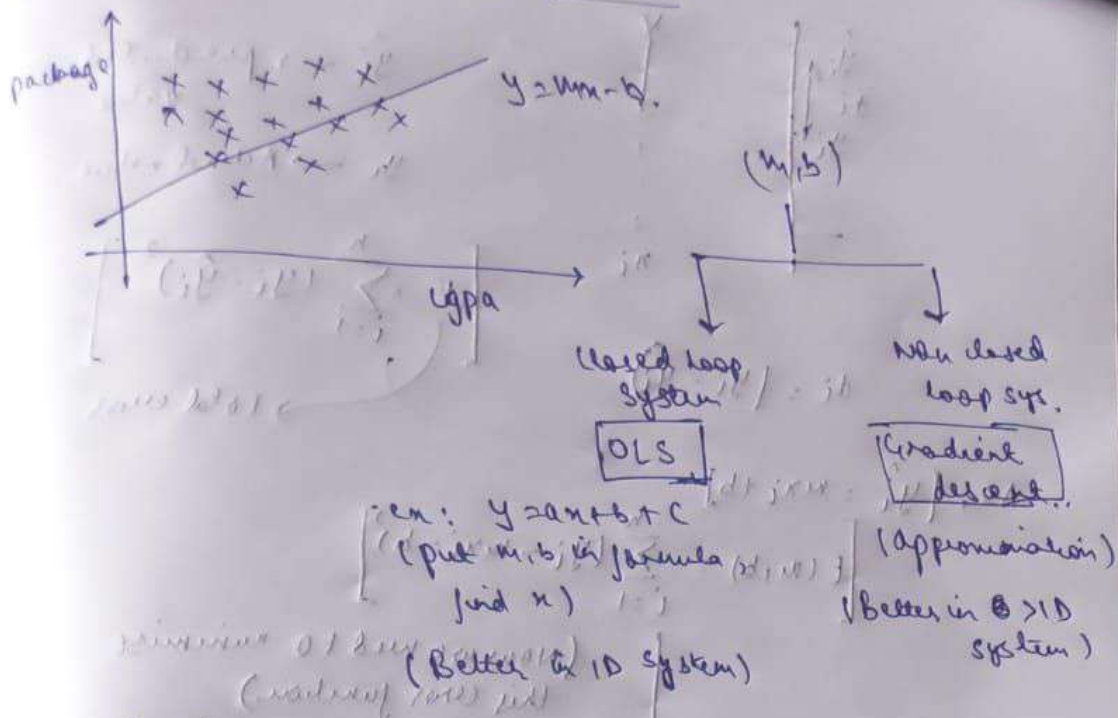


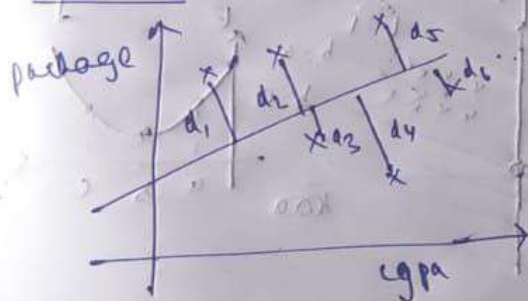
gradient descent (linear regression)



Gradient descent:

$$b = \bar{y} - m\bar{x} \quad , \quad m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Derivation



Assume n points in a 2D plane. For true values, point should lie on the line, but there are errors.

$E = d_1 + d_2 + \dots + d_n$

as $E > 0$, $(E = d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2)$

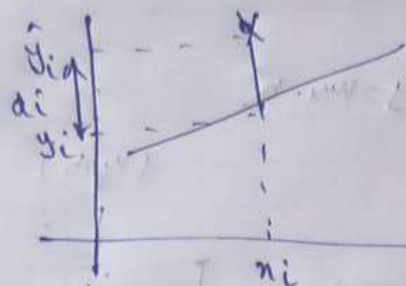
* Why not $E = |d_1| + |d_2| + \dots + |d_n|$

$$E = \sum_{i=1}^n d_i^2$$

does not remove outlier

$|x|$ is not diff at $x=0$





$\hat{y}_i \rightarrow$ predicted value
 $y_i \rightarrow$ actual value

$$d_i = |y_i - \hat{y}_i|$$

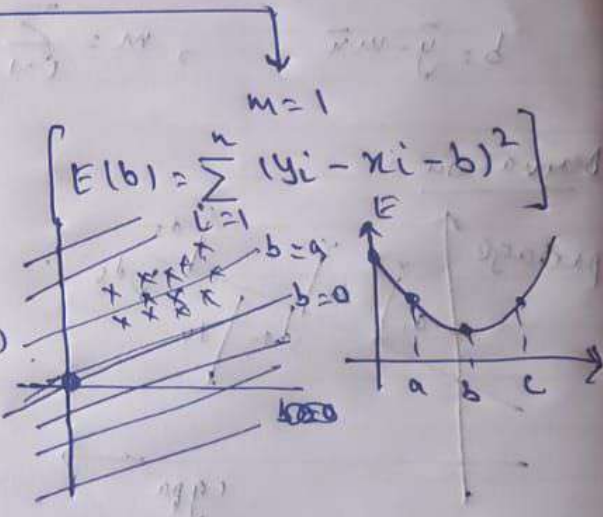
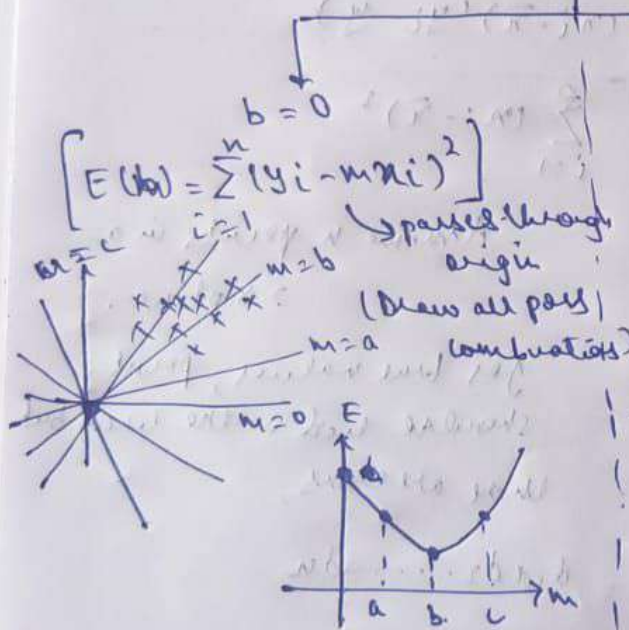
$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

total error

As, $[y_i = mx_i + b]$

$$E(m, b) = \sum_{i=1}^n (y_i - mx_i - b)^2$$

(Now we need to minimise this error function.)



for min value

$$\left(\frac{\partial E(m, b)}{\partial m} = 0 \right)$$

as 2 variables,

$$\frac{\partial E(b)}{\partial b} = 0, \frac{\partial E(m)}{\partial m} = 0$$

solve eqn,

$m, b = ?$

$$E = \sum_{i=1}^n (y_i - mx_i - b)^2$$

$$\frac{\partial E}{\partial b} = \frac{\partial}{\partial b} \left(\sum_{i=1}^n (y_i - mx_i - b)^2 \right) = \sum_{i=1}^n \frac{\partial}{\partial b} (y_i - mx_i - b)^2$$

$$= \sum_{i=1}^n 2(y_i - mx_i - b)(-1) = 0$$

$$\Rightarrow \frac{1}{n} \left[\sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - \sum_{i=1}^n b \right] = \frac{0}{n}$$

$$\Rightarrow \sum \frac{y_i}{n} - m \sum \frac{x_i}{n} - \sum \frac{b}{n} = 0$$

$$\Rightarrow \bar{y} - m\bar{x} - b = 0 \Rightarrow \cancel{b = \bar{y} - m\bar{x}}$$

$$\boxed{b = \bar{y} - m\bar{x}}$$

$$\frac{\partial E}{\partial m} = 0 \Rightarrow \sum_{i=1}^n \frac{\partial}{\partial m} (y_i - mx_i - b)^2 = 0$$

$$= \sum_{i=1}^n 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\sum 2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\sum \frac{\partial}{\partial m} (y_i - mx_i - \bar{y} + m\bar{x})^2 = 0$$

$$2(y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) = 0$$

$$\sum (y_i - \bar{y})(x_i - \bar{x}) - \sum m(x_i - \bar{x})^2 = 0$$

$$\boxed{m = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$